NEXT-GEN DBMS ARCHITECTURE

Intelligent Payroll for a Smarter Workforce

Unifies 45 employee life dimensions into 5 master domains. Powered by per-second salary streaming, semantic RAG policy search, PostgreSQL 18.4 transactions, and local MongoDB audit ledgers.

Bcrypt + JWT RBAC | Live Streaming Vault | Vector RAG Policy | Kafka Event Bus

Explore 5 Unified Domains →



Key Core Architecture Features

JWT & RBAC
Bcrypt salted hashes & signed JWT access gating 5 distinct employee roles.

Real-Time Streaming
Salary accrues per second with one-tap instant UPI wage withdrawals.

Semantic RAG
Cosine similarity vector search over labour codes and company policies.

Kafka Events
Decoupled async topics delivering real-time WebSockets notifications.

Polyglot DB
PostgreSQL 18.4 ACID core paired with MongoDB v8.3 immutable audit logs.

What-If Simulator
Monte Carlo life modeling for home loans, relocations, and tax regimes.

End-to-End Employee Lifecycle & Payroll Workflow

1 Bio-Auth → 2 Role RBAC → 3 PostgreSQL ACID → 4 Gross-to-Net → 5 Kafka Publish → 6 Mongo Audit → 7 SHA-256 Payslip

Employee Domain & Database Previews

EMPLOYEE WORKPLACE & PORTAL — DOMAIN 1



Attendance & Biometrics	99.8% Sync
On-Demand Wage Access	Auto-Approved

POSTGRESQL 18 • PAYSPLIP — pgAdmin 4

Basic Salary	₹75,000
House Rent (HRA)	₹37,500
AI Tooling Allowance	+₹10,000
Patent Royalties	+₹18,500
Net Take-Home	₹1,48,800

MONGODB v8.3 & KAFKA AI — NoSQL & Events

Audit Action	PAYROLL_DISBURSED
Audit Signature	SHA-256 Valid
Semantic RAG Policy	Cosine: 0.516
Kafka Offset	Topic #payroll-events
WebSocket Stream	• Connected

Core Technology Stack

React 19 (Frontend) | Three.js (3D Graphics) | FastAPI (Python API) | PostgreSQL (ACID Core) | MongoDB (NoSQL Audit)

Tools & Infrastructure

pgAdmin 4 (SQL Manager) | Compass (Mongo UI) | Git & GitHub (Repository) | VS Code (Environment)